

Taha Rawjani

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EDUCATION

University of Pennsylvania | B.S.E. Computer Science with EE Minor | 4.0 GPA **Expected Spring 2029**
• Relevant Coursework: Multivariable Calculus, Statistics, Physics: Mechanics, Discrete Math, Intro to Algorithms, Game Theory

EXPERIENCE

Penn Labs | Platform & DevOps Engineer **September 2026 – Present**
• Dockerized a Flask + Redis app with multi-stage builds, Redis ACL secrets, GitHub Actions CI, and Kubernetes (kind) manifests.

Penn Aerial Robotics | Embedded Software Engineer **August 2026 – Present**
• Built a ROS 2 vision pipeline detecting shapes on textured terrain via local plane-fit RMSE, with 3D positions from camera intrinsics.

Blue Vigil | Embedded Software Engineer **April 2026 – Aug 2026**
• Wrote embedded C++ for a tethered drone light system across Teensy, Raspberry Pi, 8-bit AVR, and Cube microcontrollers.
• Implemented a RS485-powered star comms network on ground and drone subsystems. Built custom ALEDMessage serialization protocol to transmit relayed messages and communicate state machine statuses through UDP, I2C, RS485, and UART devices.
• Refactored multi-board firmware monorepo into modular architecture, adding safety monitors, fault handlers, and CI/CD pipelines.
• Developed an open-source Over-The-Air (ota) remote update software and bootloader for Teensy4.1 and ATmega boards.
• Designed and fabricated a custom AVR-powered spool controller PCB in KiCad and wrote bootstrap firmware for 4+ similar AVR boards.

Maximus | Software Engineering Intern **Jun 2025 – Aug 2025**
• Created an AWS pipeline to classify email intents, reroute emails, and autonomously respond with AI for US Government Contact Centers
• Used AWS Connect, Lambda, Bedrock, EC2, Cloudwatch, Kinesis, and other services to build a scalable serverless architecture.
• Pitched to Maximus CTO, handed off and documented project to Medicare/Medicaid team for deployment. Reduced resolve times by 25%

Blueprint | Software Engineering Intern **April 2025 – June 2025**
• Automated social media (TikTok and IG) marketing with AI. Used Supabase, Next.js, Langgraph, and HeyGen avatar API.
• Built product URL webscraper in playwright & video generation API backend using remotion video creator.
• Recieved Y-Combinator Interview, hit 2000+ users peak, \$3,000+ MRR in 2 months, accepted to Founder's Inc Off-season

AWARDS & HONORS

Competitive Programming & Hackathons
• Codeforces CM and USACO Platinum Division. Team USA for International Informatics Olympiad in Teams (IIOT) 2025
• 1st place team (out of 6000+) in Category, 4th team in USA, National Cyber League Team Round Fall 2025
• 2nd place HackMIT in Sustainability 2026, 1st place HackMIT Arduino track, 2nd place HackMIT Hackster.io challenge
• 1st place at UPenn PClassic 2024; 2nd place at VT HSPC 2024; 3rd at UVA HSPC 2026; 2nd at UMD HPSC 2025
• 1st place 2023, Best Game 2024, and 2nd place 2025 at ACLHacks; Education track winner PatriotHacks; 1st place PhoenixHacks 2024

Research Awards
• Davidson Gifted Fellow in Technology for 2026 (\$25,000 scholarship). 1 of 20 nationwide selected out of 1000+ entries.
• International Science and Engineering Fair (ISEF) 2025 4th place grand & special award at in Robotics & Intelligent Machines.
• Recognized in Congressional Record (H1227) by Rep. Suhas Subramanyam for Whale Conservation Research.

PUBLICATIONS & PATENTS

T. Rawjani, M. Li. “REAL-TIME MARINE MAMMAL IDENTIFICATION AND LOCALIZATION”: USPTO No. 19/720,276, filed 2026.
T. Rawjani, M. Li. “MobyGlobal: Real-Time Whale Detection Network.” *OSM 2026*, February 2026, Glasgow, Scotland. E-Lightning.
T. Rawjani, M. Li. “MobyGlobal: A real-time whale tracking network” *Intl. Conference on Marine Science and Eng (IOCMSE) 2025*

PROJECTS

MobyGlobal **Jan 2025 - Present**
• Designed Moby, a two-branch CBAM + CNN-BiLSTM ensemble for whale call detection, trained on 30k+ clips (Cornell, Watkins, NOAA).
• Achieved 0.977 AUROC with 344K parameters, outperforming ResNet, VGG, and EfficientNet and Cornell's 0.72 benchmark.
• Built a real-time REST pipeline running sliding-window inference on buoy audio: 26 ms API response, <0.5 s per update.
• Built a \$442 solar ESP32 buoy (72 hrs off-grid, Von Mises tested) with a \$40 custom hydrophone, 2x cheaper than Aquarian.

Neo OS **Jul. 2022 - Aug. 2023**
• An operating system written from scratch in C++ and assembly with a UNIX-based virtual file system, and multithreading support.
• Wrote TahaScript, a dynamically interpreted scripting language; and TWM, a customizable window manager built for Neo-OS.
• Programmed an in-house image format library (.nic) with 24-bit color support alongside custom font and VGA drivers
• Created drivers for hard drives (AHCI), floppy disks (FDC), timers (PIT, APIC), interfaces (PS/2 mouse/kbd, PCI), filesystems (FAT32)

6502 GPT **June 2026 - July 2026**
• Built a Ben-Eater inspired 6502 breadboard computer using a 8-bit W65C02S MPU, 32k ROM/RAM, and a 555-based clock (65 kHz).
• Programmed an On-Chip debugger & EEPROM flasher, tracing 6502 clock, stack, registers, interrupts, and opcodes w/ Arduino Mega.
• Wrote an OS with vasm-oldstyle and cc65 cross compilers, supporting I2C and serial communication with LCDs, SD storage, and keypads.
• Ported custom C-based deep learning library (CMLIB) and ran an in-house small-paramter LLM on 4 MHz clock speed.

TECHNICAL SKILLS

Electrical & Embedded: Embedded C/C++, Microcontrollers (Arduino, ESP32, Teensy4.1), MPUs (6502, AVR), Raspberry Pi, HV power systems and rectifiers, multi-board firmware architecture, Fusion 360, 3D printing, KiCad PCB Design, Circuit Analysis

Software Engineering & Programming: C++, C, Python, Rust, TypeScript/JavaScript, x86 & 6502 assembly, Git, AWS (Lambda, Bedrock, EC2, Connect, Kinesis, CloudWatch), Supabase, Next.js, REST APIs, web scraping, operating systems (device drivers, virtual file systems, multithreading), protocol design, language & interpreter design

Machine Learning: PyTorch, TensorFlow, ensemble learning, attention mechanisms, acoustic signal classification, intent classification, LLM development, agent orchestration (LangGraph, MCP)